

FE5108 Midterm: Who Does What

5 people (A to E) | Due Sunday 18 October 2026, 23:59 Singapore time

The project in one line: pick 25 stocks, then check with real data whether the textbook theory (best portfolio, CAPM, factor models) actually holds. Report max 12 pages, plus code that runs.

Who does what

Person A: Data and code

You get the data ready so everyone else can start, and at the end you make sure all the code runs.

What to do

1. Download monthly prices for the 25 stocks and the market index. Save the raw files in **data/raw/** and never edit them.
2. Download the risk-free rate and the factor data (Ken French website).
3. Turn prices into monthly returns. Fix missing values. Write down every cleaning choice you make.
4. Save one clean returns file in **data/clean/**. Everyone reads from this file only.
5. Send the group a short note: returns in decimals or percent, monthly or yearly numbers, chart colours and font size.
6. At the end: make **run_all.py** run everyone's code in order, and write the README (data files, package versions, how to run).
7. Run everything from zero on a clean computer to confirm it works.

What you hand in

- Clean returns file: by **Fri 9 Oct**
- run_all.py and README, tested: by **Fri 16 Oct**

Person B: Stage 1, the best portfolio

You compute the "best" portfolio from the data and show that it looks nothing like the real market.

What to do

1. Make a table of basic numbers for each stock: average return, volatility, and so on.
2. Compute the tangency weights (the mathematically best mix of the 25 stocks).
3. Compute the value weights (each stock's market cap divided by the total market cap of the 25).
4. Draw **one** chart with both sets of weights side by side.
5. Split the sample into two halves and recompute the tangency weights for each half. Show how much they change.
6. Recompute once more with no short-selling allowed, to show the weights become more reasonable.
7. Write about 2.5 pages: why the two sets of weights disagree, and what this says about estimation error.

What you hand in

- Table and charts: by **Mon 12 Oct**
- Written section: by **Thu 15 Oct**

Person C: Stage 2, testing CAPM

You check whether a stock's beta really explains its average return.

What to do

1. For each stock, run one regression: stock excess return on market excess return. That is 25 regressions.
2. Put the results in one table: beta, alpha, t-statistic of alpha, R-squared.
3. Draw a scatter plot: beta on the x-axis, average excess return on the y-axis, one dot per stock.
4. Add two lines to the plot: the line fitted through the dots, and the line CAPM predicts.
5. Count how many stocks have $|t|$ above 2, and compare with the 1 or 2 you would expect from luck alone. (Optional: also run the GRS test.)
6. Compare the two lines: is the fitted one flatter? Does it start above zero? Explain why.
7. Write about 2.5 pages. Keep your code tidy, because E will reuse it.

What you hand in

- Table and plot: by **Mon 12 Oct**
- Written section: by **Thu 15 Oct**

Person D: Stage 3, adding factors

You check whether adding size, value or momentum fixes what CAPM got wrong.

What to do

1. Choose at least two extra factors (for example size and value) from the data A downloaded.
2. Write one short paragraph per factor: what it is and how it is built.
3. Re-run the 25 regressions with the market **plus** the extra factors.
4. Make one table or chart comparing each stock's alpha before (from C) and after.
5. Write about 2.5 pages: which of C's problems went away, and which are still there.

What you hand in

- Table or chart: by **Mon 12 Oct**
- Written section: by **Thu 15 Oct**

Person E: Stage 4 and the final report

You write the final recommendation and turn five people's writing into one clean report.

What to do

1. Now: write a first rough version of the memo based on what you expect to find. Set up the report template.
2. Re-run C's regressions using a different market index (for example MSCI World). Show how much the betas and alphas change.
3. Write the memo (max 2 pages) answering three questions: (a) should an investor hold the index, the tangency portfolio, or something in between? (b) what does changing the market index tell us about testing CAPM? (c) which of our results do we trust least, and why?
4. Write the 1-page summary for the front. It states what we found, not what we did.
5. Put all sections together: same wording, same number format, same chart style, and 12 pages or fewer.
6. Check that every chart title says its conclusion in one sentence.
7. Put together the appendix: who did what (signed by all five), and one sentence on which AI tools were used and for what.

What you hand in

- Memo draft: by **Thu 15 Oct**
- Full report for proofreading: by **Fri 16 Oct**

Calendar

When	What happens
Wed 7 Oct	Everyone: agree on the 25 stocks, the time period, the market index and the risk-free rate.
Thu 8 - Fri 9 Oct	A finishes the data. B, C, D start coding using the backup data on Canvas. E writes the rough memo.
Sat 10 - Mon 12 Oct	B, C, D produce their tables and charts with the real data.
Mon 12 Oct, evening	Meeting 1: everyone shows their results. Check the numbers agree with each other.
Tue 13 - Wed 14 Oct	B, C, D write their sections. E re-runs Stage 2 with the second index. A starts run_all.py.
Thu 15 Oct	All writing goes to E. Meeting 2: agree on the conclusions and the recommendation.
Fri 16 Oct	E finishes the full report. A tests all code from zero.
Sat 17 Oct	Everyone reads the whole report, fixes mistakes, and signs the contribution statement.
Sun 18 Oct	Spare day. Send one email with the PDF report and the code before 23:59.

Rules for everyone

- Read data only from **data/clean/**. Do not download your own copy.
- Save every chart and table to **output/**.
- Only edit your own code file and your own report section.
- Run **git pull** before you start working each time.
- Do only what the handout asks for. No extra charts unless they change a conclusion.